

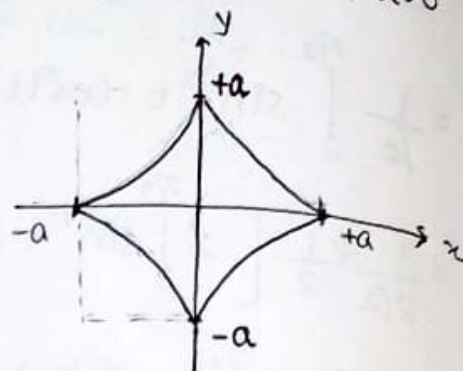
TUTORIAL 5:-

① Find total perimeter of curve asteroïd $x^{2/3} + y^{2/3} = a^{2/3}$, $a > 0$

Soln:- Given $x^{2/3} + y^{2/3} = a^{2/3}$ — ①

Total perimeter = 4 × Length of curve
in Ist Quad

$$\text{Total perimeter} = 4 \times \int_0^a \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$



Differentiating ① w.r.t x ,

$$\frac{2}{3} x^{-1/3} + \frac{2}{3} y^{-1/3} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-x^{-1/3}}{y^{-1/3}} = -\frac{y^{1/3}}{x^{1/3}}$$

$$\therefore \text{Total perimeter} = 4 \int_0^a \sqrt{1 + \left(-\frac{y^{1/3}}{x^{1/3}}\right)^2} dx$$

$$= 4 \int_0^a \sqrt{\frac{x^{2/3} + y^{2/3}}{x^{2/3}}} dx$$

$$= 4 \int_0^a \sqrt{\frac{a^{2/3}}{x^{2/3}}} dx \quad (\text{from ①})$$

$$= 4 \int_0^a \frac{a^{1/3}}{x^{1/3}} dx$$

$$= 4a^{1/3} \int_0^a x^{-1/3} dx$$

$$= 4a^{1/3} \left(\frac{x^{2/3}}{2/3} \right)_0^a$$

$$= 6a^{1/3} \cdot a^{2/3}$$

$$\boxed{\text{Perimeter} = 6a}$$

Q Find the length of the curve $3x^2=y^3$ between $y=0$ and $y=1$

Soln: Given $3x^2-y^3=0$ - ①

We know that,

$$\text{Length of curve} = \int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

Differentiating ① w.r.t 'y',

$$6x \frac{dx}{dy} - 3y^2 = 0$$

$$\frac{dx}{dy} = \frac{3y^2}{6x}$$

$$\frac{dx}{dy} = \frac{y^2}{2x}$$

$$\therefore \text{Length} = \int_0^1 \sqrt{1 + \left(\frac{y^2}{2x}\right)^2} dy$$

$$= \int_0^1 \sqrt{\frac{4x^2 + y^4}{4x^2}} dy$$

$$= \int_0^1 \sqrt{\frac{4y^3 + 3y^4}{4y^3}} dy \quad (\text{from ①})$$

$$= \int_0^1 \sqrt{\frac{4+3y}{4}} dy$$

$$= \int_0^1 \left(1 + \frac{3y}{4}\right)^{1/2} dy$$

$$= \frac{1}{2} \int_0^1 \sqrt{3y+4} dy$$

$$= \frac{1}{2} \left[\frac{(3y+4)^{3/2}}{\frac{3}{2} \cdot (3)} \right]_0^1$$

$$\text{Length} = \frac{1}{9} (7^{3/2} - 8)$$

③ Find the area common to the circles $r = a \cos \theta$ & $r = a \sin \theta$.

Soln: Given $r = a \cos \theta$ — (1)
 $r = a \sin \theta$ — (2)

We know that,

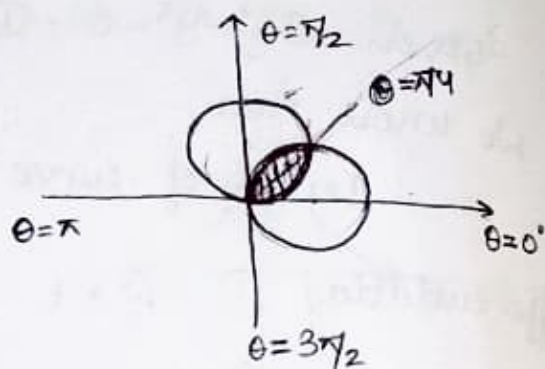
$$\text{Area} = \frac{1}{2} \int_{\theta_1}^{\theta_2} r^2 d\theta$$

for point of intersection,

$$r = a \cos \theta = a \sin \theta$$

$$\tan \theta = 1$$

$$\boxed{\theta = \pi/4} \text{ — (3)}$$



$$\begin{aligned} \therefore \text{Area} &= \frac{1}{2} \int_0^{\pi/4} a^2 \sin^2 \theta d\theta + \frac{1}{2} \int_{\pi/4}^{\pi/2} a^2 \cos^2 \theta d\theta \\ &= \frac{a^2}{2} \int_0^{\pi/4} \frac{1 - \cos 2\theta}{2} d\theta + \frac{a^2}{2} \int_{\pi/4}^{\pi/2} \frac{1 + \cos 2\theta}{2} d\theta \\ &= \frac{a^2}{4} \int_0^{\pi/4} 1 - \cos 2\theta d\theta + \frac{a^2}{4} \int_{\pi/4}^{\pi/2} 1 + \cos 2\theta d\theta \\ &= \frac{a^2}{4} \left(\theta - \frac{\sin 2\theta}{2} \right)_0^{\pi/4} + \frac{a^2}{4} \left(\theta + \frac{\sin 2\theta}{2} \right)_{\pi/4}^{\pi/2} \\ &= \frac{a^2}{4} \left(\frac{\pi}{4} - \frac{1}{2} - 0 \right) + \frac{a^2}{4} \left[\frac{\pi}{2} + 0 - \left(\frac{\pi}{4} + \frac{1}{2} \right) \right] \\ &= \frac{a^2}{4} \left(\frac{\pi}{4} - \frac{1}{2} \right) + \frac{a^2}{4} \left(\frac{\pi}{4} - \frac{1}{2} \right) \end{aligned}$$

$$\boxed{\text{Area} = \frac{a^2}{4} \left(\frac{\pi}{2} - 1 \right)}$$

Q) Find the surface area of solid obtained when cardioid $r = a(1 + \cos\theta)$, $a > 0$ is rotated about the initial line.

Soln: Given curve $r = a(1 + \cos\theta)$ — (1)

We know that

$$\text{surface area } S = 2\pi \int_{\theta_1}^{\theta_2} r \sin\theta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

Here θ varies from $\theta = 0$ to $\theta = \pi$

Differentiating (1) wrt θ , we get,

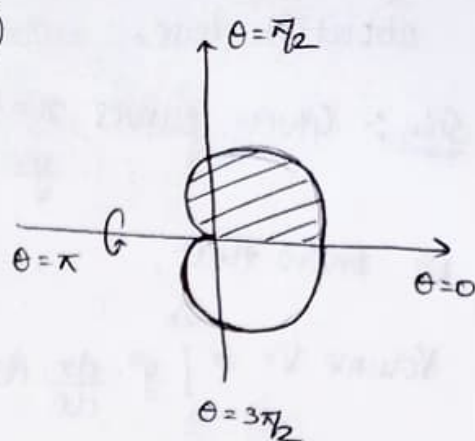
$$\frac{dr}{d\theta} = -a \sin\theta$$

$$\therefore \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} = \sqrt{a^2(1 + \cos\theta)^2 + a^2 \sin^2\theta}$$

$$= a \sqrt{1 + \cos^2\theta + 2\cos\theta + \sin^2\theta}$$

$$= a \sqrt{2 + 2\cos\theta}$$

$$= 2a \cos\frac{\theta}{2} \quad \because (1 + \cos A = 2\cos^2\frac{A}{2})$$



$$S = 2\pi \int_0^{\pi} r \sin\theta \cdot 2a \cos\frac{\theta}{2} d\theta$$

$$S = 4\pi a \int_0^{\pi} a(1 + \cos\theta) \sin\theta \cos\frac{\theta}{2} d\theta$$

$$S = 4\pi a^2 \int_0^{\pi} 2\cos^2\frac{\theta}{2} \cdot 2\sin\frac{\theta}{2} \cos\frac{\theta}{2} \cdot \cos\frac{\theta}{2} d\theta$$

$$S = 16\pi a^2 \int_0^{\pi} \sin\frac{\theta}{2} \cos^4\frac{\theta}{2} d\theta$$

$$\because (\sin 2A = 2\sin A \cos A)$$

Put $\frac{\theta}{2} = t \rightarrow$ for $\theta = 0, t = 0$
 $\theta = \pi, t = \frac{\pi}{2}$
 $d\theta = 2dt$

$$S = 16\pi a^2 \int_0^{\pi/2} \sin(t) \cdot \cos^4(t) \cdot 2dt$$

$$S = 32\pi a^2 \int_0^{\pi/2} \sin(t) \cos^4(t) dt$$

$$S = 32\pi a^2 \cdot \frac{(1)(4-1)(4-3)}{(5)(5-2)(5-4)(1)}$$

$$S = 32\pi a^2 \cdot \frac{3 \times 1}{5 \times 3 \times 1 \times 1}$$

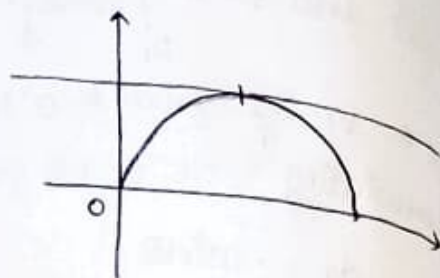
$$\boxed{S = \frac{32\pi a^2}{5}}$$

⑤ Find the volume of the solid generated by the revolution of the cycloid $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$, $a > 0$, $0 \leq \theta \leq 2\pi$ about its base.

Soln:- Given curves $x = a(\theta - \sin \theta)$ - ①
 $y = a(1 - \cos \theta)$ - ②

We know that,

$$\text{Volume } V = \pi \int_0^{2\pi} y^2 \cdot \frac{dx}{d\theta} \cdot d\theta$$



$$V = \pi \int_0^{2\pi} a^2(1 - \cos \theta)^2 \cdot a(1 - \cos \theta) \cdot d\theta$$

$$V = \pi a^3 \int_0^{2\pi} (1 - \cos \theta)^3 d\theta$$

$$V = \pi a^3 \int_0^{2\pi} (2 \sin^2 \theta/2)^3 d\theta \quad [\because 1 - \cos A = 2 \sin^2 A/2]$$

$$V = 8\pi a^3 \int_0^{2\pi} \sin^6 \theta/2 d\theta$$

Put $\frac{\theta}{2} = t \rightarrow$ for $\theta = 0, t = 0$

$d\theta = 2dt$ $\theta = 2\pi, t = \pi$

$$V = 16\pi a^3 \int_0^{\pi} \sin^6(t) dt$$

$$V = 16\pi a^3 \int_0^{\pi} 2^6 \cdot \sin^6(t/2) \cos^6(t/2) dt \quad (\because \sin 2A = 2 \sin A \cos A)$$

$$V = 2^{10} \pi a^3 \int_0^{\pi} \sin^6(t/2) \cos^6(t/2) dt$$

Put $\frac{t}{2} = w \rightarrow$ for $t = 0, w = 0$

$dt = 2dw$ $t = \pi, w = \pi/2$

$$V = 2^{10} \pi a^3 \int_0^{\pi/2} \sin^6(w) \cos^6(w) dw \Rightarrow V = 2^{10} \pi a^3 \frac{(5 \times 3 \times 1)(5 \times 3 \times 1)}{12 \times 10 \times 8 \times 6 \times 4 \times 2 \times 1} \left(\frac{\pi}{2}\right)$$

$$\boxed{V = 5\pi^2 a^3}$$